

Lecture 04: Distances Between Probability Measures

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Consider the simple scenario: you roll two six-sided dice, one fair and one slightly biased so that it lands on "6" more often. Without a way to compute the distance between their outcome distributions, you would only know qualitatively that the biased die "feels" different. By measuring the distance, however, you can say precisely that the biased die's distribution is, for instance, 0.5 "units" away from uniform, which immediately tells you how much more likely it is to roll a 6 compared to the fair die. This quantitative gap is essential whenever you need to decide if two processes are "close" or whether a detected shift is statistically significant and warrants further action.

1 Definitions

We first briefly recall the definitions of a probability measure and a metric, which will be needed in what follows.

Definition 1 (Probability measure). Let $(\Omega, \mathcal{F})$ be a measurable space. A function

$$\mathbb{P} : \mathcal{F} \longrightarrow [0, 1]$$

is called a probability measure if it satisfies the following axioms:

1. $\mathbb{P}(\emptyset) = 0$.
2. $\mathbb{P}(\Omega) = 1$.
3. **Countable additivity:** For any countable collection $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{F}$ of pairwise disjoint sets,

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Examples Consider a single roll of a fair six-sided die. The sample space is $\Omega = \{1, 2, 3, 4, 5, 6\}$. The σ -algebra: $\mathcal{F}$ is the full power set $\mathcal{F} = 2^{\Omega}$, i.e. all subsets of $\{1, 2, 3, 4, 5, 6\}$. For each elementary outcome $i \in \Omega$, the probability measure is $\mathbb{P}(\{i\}) = \frac{1}{6}$.

Another well-known example is the Lebesgue measure of an interval $[a, b]$ which is simply its length $m([a, b]) = b - a$.

Definition 2 (Metric). Let X be a nonempty set. A function

$$d : X \times X \longrightarrow [0, \infty)$$

is called a metric (or distance function) on X if for all $x, y, z \in X$ the following conditions hold:

1. **Non-negativity:** $d(x, y) \geq 0$.
2. **Identity of indiscernibles:** $d(x, y) = 0 \iff x = y$.
3. **Symmetry:** $d(x, y) = d(y, x)$.
4. **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

If d satisfies these axioms, the pair (X, d) is called a metric space.

2 Divergences and Distances Between Probability Measures

Let $(\mathcal{X}, \mathcal{F})$ be a measurable space, and let $\mathbb{P}$ and $\mathbb{Q}$ be probability measures on this space. Let $p = \frac{d\mathbb{P}}{dx}$ and $q = \frac{d\mathbb{Q}}{dx}$ denote the respective densities with respect to a common dominating measure dx (e.g., the Lebesgue measure or counting measure).

We will use the notation $\mathbb{P} \ll \mathbb{Q}$ which means that the measure $\mathbb{P}$ is **absolutely continuous** with respect to the measure $\mathbb{Q}$, as defined next.

Definition 3 (Absolute continuity). For two measures $\mathbb{P}, \mathbb{Q}$ defined on the same measurable space $(\Omega, \mathcal{F})$:

$$\mathbb{P} \ll \mathbb{Q} \iff \forall A \in \mathcal{F}, \mathbb{Q}(A) = 0 \Rightarrow \mathbb{P}(A) = 0.$$

That is, whenever $\mathbb{Q}$ assigns zero measure to a set, $\mathbb{P}$ also assigns zero measure to it.

Kullback–Leibler divergence We begin with the Kullback–Leibler divergence, one of the most widely used divergences in data science, because it provides a principled way to measure how one probability distribution differs from another and is closely connected to maximum likelihood estimation, information theory, and many machine learning objectives. However, one will see from its definition that it is not a metric as it does not satisfy the symmetric property.

Definition 4 (Kullback–Leibler divergence). If $\mathbb{P} \ll \mathbb{Q}$, the Kullback–Leibler (KL) divergence is defined as

$$\text{KL}(\mathbb{P} \parallel \mathbb{Q}) = \int_{\mathcal{X}} p(x) \log \left(\frac{p(x)}{q(x)} \right) dx,$$

with the conventions that $0 \log \left(\frac{0}{q(x)} \right) = 0$ and $p(x) \frac{\log(p(x))}{0} = \infty$ for $p(x) > 0$.

Let's go back to the example of two dices discussed at the beginning of these notes. Consider the sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$ with two dice distributions:

$$\mathbb{Q}(i) = \frac{1}{6}, \quad i = 1, \dots, 6, \quad \mathbb{P}(i) = \begin{cases} \frac{2}{15}, & i = 1, \dots, 5, \\ \frac{1}{3}, & i = 6. \end{cases}$$

The Kullback–Leibler divergence of $\mathbb{P}$ from $\mathbb{Q}$ is

$$\begin{aligned} \text{KL}(\mathbb{P} \parallel \mathbb{Q}) &= \sum_{i=1}^6 P(i) \ln \left(\frac{P(i)}{Q(i)} \right) \\ &= \sum_{i=1}^5 \frac{2}{15} \ln \left(\frac{\frac{2}{15}}{\frac{1}{6}} \right) + \frac{1}{3} \ln \left(\frac{\frac{1}{3}}{\frac{1}{6}} \right) \\ &= \frac{2}{3} \ln \left(\frac{4}{5} \right) + \frac{1}{3} \ln(2) \\ &\approx 0.0823 \text{ nats.} \end{aligned}$$

As another example, let's consider the distance between two Gaussian distributions illustrated in Fig. 1 where one can observe that the area being integrated for $\text{KL}(\mathbb{P} \parallel \mathbb{Q})$ is on the support of the distribution $\mathbb{P}$.

Jensen–Shannon divergence We next turn to the Jensen–Shannon divergence, a widely used divergence that symmetrizes the Kullback–Leibler divergence. Note that it is also not a metric because it does not satisfy the triangle inequality.

Definition 5 (Jensen–Shannon divergence). Let $\mathbb{M} = \frac{1}{2}(\mathbb{P} + \mathbb{Q})$. The Jensen–Shannon (JS) divergence is defined as

$$\text{JS}(\mathbb{P} \parallel \mathbb{Q}) = \frac{1}{2} \text{KL}(\mathbb{P} \parallel \mathbb{M}) + \frac{1}{2} \text{KL}(\mathbb{Q} \parallel \mathbb{M}).$$

It is symmetric and always finite. Moreover, the square root $\sqrt{\text{JS}}$ defines a metric.

Total variation distance Finally, we introduce the total variation (TV) distance which is a proper metric on the space of probability measures (exercise: try to prove it satisfies all the required properties).

Definition 6 (Total variation distance). The total variation (TV) distance is defined as

$$\text{TV}(\mathbb{P}, \mathbb{Q}) = \sup_{A \in \mathcal{F}} |\mathbb{P}(A) - \mathbb{Q}(A)| = \frac{1}{2} \int_{\mathcal{X}} |p(x) - q(x)| dx.$$

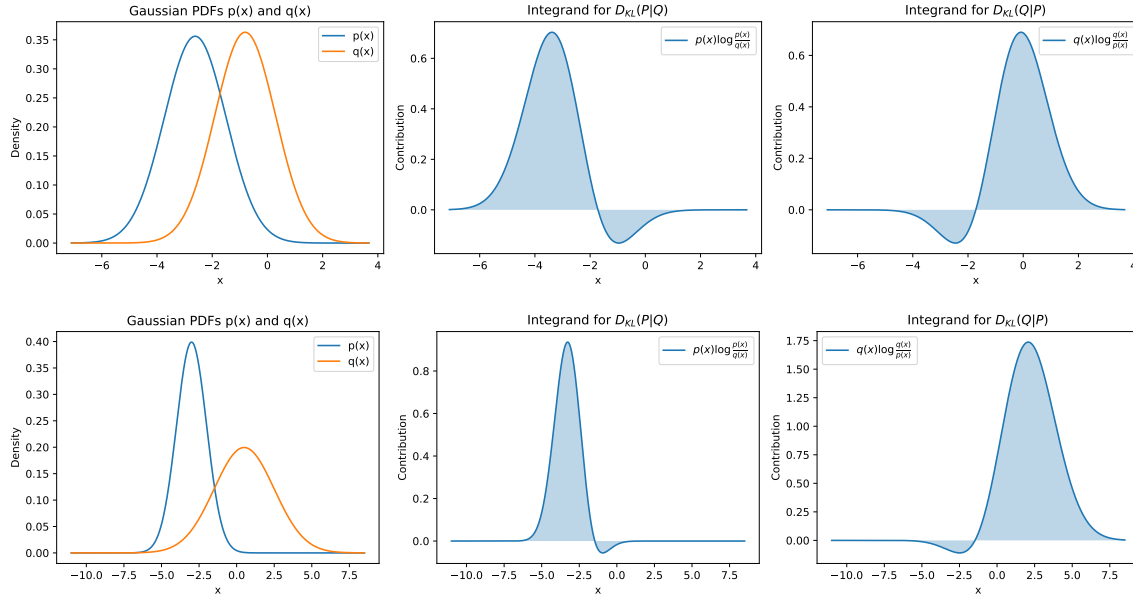


Figure 1: Examples of KL divergences computed on Gaussian distributions.

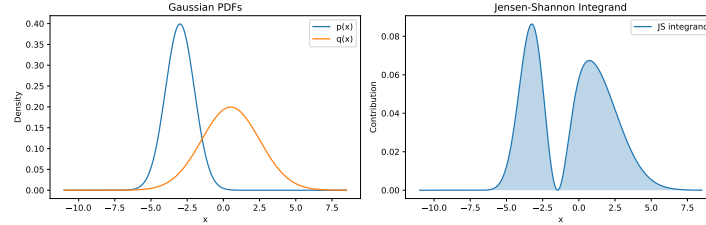


Figure 2: Illustration of JS divergence computed between two Gaussians.

The following inequality (known as Pinsker's inequality) bounds the total variation distance in terms of the Kullback–Leibler divergence.

Theorem 1 (Pinsker's Inequality). *Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures on $(\mathcal{X}, \mathcal{F})$ which are both absolutely continuous with respect to some reference measure μ , with densities $p = d\mathbb{P}/d\mu$ and $q = d\mathbb{Q}/d\mu$. Then*

$$D_{\text{KL}}(\mathbb{P}||\mathbb{Q}) = \int_{\mathcal{X}} p(x) \ln \frac{p(x)}{q(x)} d\mu(x) \geq 2 \text{TV}(\mathbb{P}, \mathbb{Q})^2,$$

where

$$\text{TV}(\mathbb{P}, \mathbb{Q}) = \frac{1}{2} \int_{\mathcal{X}} |p(x) - q(x)| d\mu(x).$$

Proof. Set

$$r(x) = \frac{p(x)}{q(x)} - 1,$$

so that $r(x) \geq -1$ and $p(x) = q(x)(1 + r(x))$. Then

$$D_{\text{KL}}(\mathbb{P}||\mathbb{Q}) = \int_{\mathcal{X}} p(x) \ln \frac{p(x)}{q(x)} d\mu = \int_{\mathcal{X}} q(x) [(1 + r(x)) \ln(1 + r(x))] d\mu. \quad (1)$$

Next, we will slightly modify the above equation using the fact that $\int_{\mathcal{X}} q(x) r(x) d\mu(x) = 0$. This can be shown as follows. By definition,

$$r(x) = \frac{p(x)}{q(x)} - 1 \implies q(x)r(x) = p(x) - q(x).$$

Integrate both sides:

$$\int_{\mathcal{X}} q(x)r(x) d\mu(x) = \int_{\mathcal{X}} (p(x) - q(x)) d\mu(x) = \int_{\mathcal{X}} p(x) d\mu(x) - \int_{\mathcal{X}} q(x) d\mu(x).$$

But p and q are densities of probability measures $\mathbb{P}$ and $\mathbb{Q}$, so

$$\int_{\mathcal{X}} p(x) d\mu(x) = 1, \quad \int_{\mathcal{X}} q(x) d\mu(x) = 1.$$

Hence

$$\int_{\mathcal{X}} q(x)r(x) d\mu(x) = 1 - 1 = 0.$$

Back to Eq. (1), we now have

$$D_{\text{KL}}(\mathbb{P}||\mathbb{Q}) = \int_{\mathcal{X}} p(x) \ln \frac{p(x)}{q(x)} d\mu = \int_{\mathcal{X}} q(x) [(1 + r(x)) \ln(1 + r(x)) - r(x)] d\mu.$$

Using the inequality

$$(1 + r) \ln(1 + r) - r \geq \frac{1}{2} \frac{r^2}{1 + \frac{r}{3}},$$

we get

$$D_{\text{KL}}(\mathbb{P}||\mathbb{Q}) \geq \frac{1}{2} \int_{\mathcal{X}} q(x) \frac{r(x)^2}{1 + \frac{r(x)}{3}} d\mu.$$

By Titu's lemma ¹,

$$\int_{\mathcal{X}} q \frac{r^2}{1 + r/3} \geq \frac{(\int_{\mathcal{X}} q |r|)^2}{\int_{\mathcal{X}} q (1 + r/3)} = (\int_{\mathcal{X}} q |r|)^2,$$

since $\int_{\mathcal{X}} q(1 + r/3) d\mu = 1$. Hence

$$D_{\text{KL}}(\mathbb{P}||\mathbb{Q}) \geq \frac{1}{2} \left(\int_{\mathcal{X}} q(x) |r(x)| d\mu \right)^2 = \frac{1}{2} \left(\int_{\mathcal{X}} |p(x) - q(x)| d\mu \right)^2 = 2 \text{TV}(\mathbb{P}, \mathbb{Q})^2.$$

□

To illustrate the geometric relationship between total variation and Kullback–Leibler divergence, we visualize these quantities over the probability simplex in Figure 3. For a fixed reference distribution y , the plots show the level sets of $\text{TV}(x||y)$ and $\sqrt{\text{KL}(x||y)}$, as well as their ratio $\text{TV}(x||y)/\sqrt{2\text{KL}(x||y)}$. The latter highlights where Pinsker's inequality is tight or loose throughout the simplex. The reference point y is indicated by a red marker at the center of the triangle.

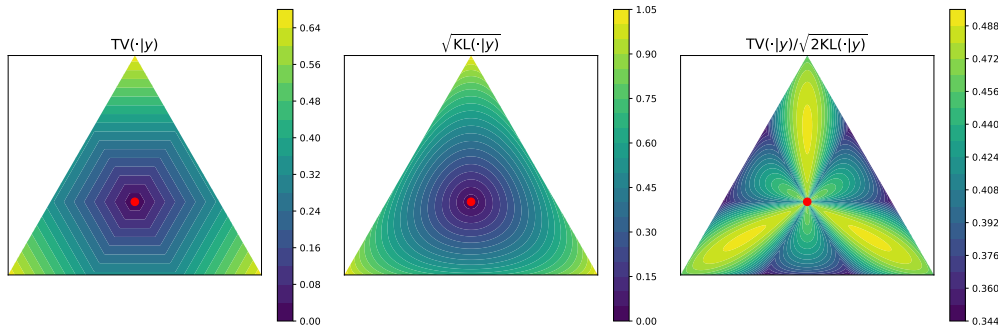


Figure 3: Comparison of the level sets of total variation, square-root KL divergence, and their ratio $\text{TV}(x||y)/\sqrt{2\text{KL}(x||y)}$ on the probability simplex. The ratio illustrates how close Pinsker's inequality $\text{TV}(x, y) \leq \sqrt{2\text{KL}(x||y)}$ is to equality across the domain.

¹Titu's Lemma states the following: Let $a_1, \dots, a_n \in \mathbb{R}$ and $b_1, \dots, b_n > 0$. Then

$$\sum_{i=1}^n \frac{a_i^2}{b_i} \geq \frac{(\sum_{i=1}^n |a_i|)^2}{\sum_{i=1}^n b_i}.$$

3 Wasserstein Distance and Kantorovich–Rubinstein Duality

3.1 Definition of Wasserstein distance

Coupling – Definition Let $\mathcal{X}$ be a finite metric space with metric $d : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$. Let μ, ν be two probability distributions on $\mathcal{X}$. Define the set of couplings as:

$$\Gamma(\mu, \nu) = \left\{ \gamma : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0} \left| \sum_y \gamma(x, y) = \mu(x), \sum_x \gamma(x, y) = \nu(y) \right. \right\},$$

where γ is a probability distribution on $\mathcal{X} \times \mathcal{X}$.

In this context, a coupling of the two probability measures μ and ν on the same finite set $\mathcal{X}$ is simply a way of ”pairing up” mass from μ with mass from ν .

Example 1: Correlated coupling Let $\mathcal{X} = \{a, b\}$ and define two marginal distributions

$$\mu(a) = \frac{1}{2}, \quad \mu(b) = \frac{1}{2}, \quad \nu(a) = \frac{1}{2}, \quad \nu(b) = \frac{1}{2}.$$

Define the “comonotone” coupling by placing all mass on the diagonal:

$$\gamma_{\text{corr}}(x, y) = \begin{cases} \frac{1}{2}, & (x, y) = (a, a) \text{ or } (b, b), \\ 0, & \text{otherwise.} \end{cases}$$

One can easily check that

$$\sum_{y \in \mathcal{X}} \gamma_{\text{corr}}(x, y) = \mu(x), \quad \sum_{x \in \mathcal{X}} \gamma_{\text{corr}}(x, y) = \nu(y),$$

so $\gamma_{\text{corr}} \in \Gamma(\mu, \nu)$. Intuitively, when you draw $x \sim \mu$, you always “pair” it with the same value $y = x$ under ν .

Example 2. Independent (product) coupling Define the product (independent) coupling by

$$\gamma_{\text{ind}}(x, y) = \mu(x) \nu(y) = \begin{cases} \frac{1}{4}, & (x, y) \in \{a, b\} \times \{a, b\}, \\ 0, & \text{otherwise.} \end{cases}$$

Again one checks

$$\sum_{y \in \mathcal{X}} \gamma_{\text{ind}}(x, y) = \mu(x) \sum_y \nu(y) = \mu(x), \quad \sum_{x \in \mathcal{X}} \gamma_{\text{ind}}(x, y) = \nu(y),$$

so $\gamma_{\text{ind}} \in \Gamma(\mu, \nu)$. Here the two samples are drawn independently.

		y	
		a	b
x	a	$\frac{1}{2}$	0
	b	0	$\frac{1}{2}$

(a) Dependent (comonotone) coupling

		y	
		a	b
x	a	$\frac{1}{4}$	$\frac{1}{4}$
	b	$\frac{1}{4}$	$\frac{1}{4}$

(b) Independent coupling

Figure 4: Two couplings of $(x, y) \in \{a, b\}^2$: dependent (left) and independent (right).

Example 3 Another example of a coupling between two 1-dimensional distribution is shown in Figure 5.

Definition of Wasserstein distance We are now ready to introduce the general definition of Wasserstein distance.

Definition 7 (Wasserstein distance). Let $\Gamma(\mu, \nu)$ denote the set of all couplings of μ and ν . For $p \geq 1$, the p -Wasserstein distance is defined as

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} \|x - y\|^p d\gamma(x, y) \right)^{1/p}.$$

A few remarks are in order:

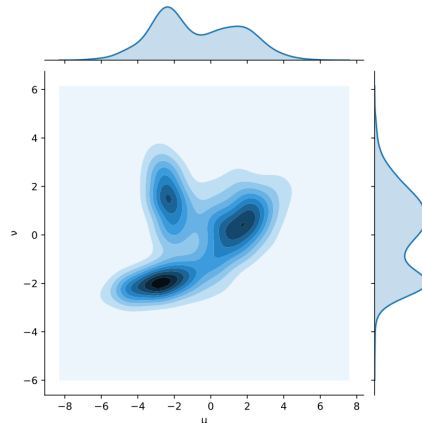


Figure 5: Two one-dimensional distributions μ and ν , plotted on the x and y axes, and one possible joint distribution that defines a transport plan between them. The joint distribution/transport plan is not unique. *Source: wikipedia.*

- Among all possible couplings/transportation plans γ that convert μ into ν , the $\inf$ ensures that we take the cheapest one.
- Note that the term $c(x, y) = \|x - y\|$ in the integral is the cost of moving one unit of mass from point x to point y .
- $\gamma(x, y)$ is the amount of mass starting at x that is sent to the destination y . Therefore

$$\int_A \int_B d\gamma(x, y) = \text{mass moved from region } A \text{ to region } B.$$

Wasserstein distance vs KL divergence Consider the following two distributions:

$$\mu = \frac{1}{2}(\delta_0 + \delta_{10}), \quad \nu = \frac{1}{2}(\delta_1 + \delta_{11}).$$

The support of μ is

$$\text{supp}(\mu) = \{0, 10\},$$

and the support of ν is

$$\text{supp}(\nu) = \{1, 11\}.$$

There is no overlap:

$$\nu(0) = 0 \quad \text{but} \quad \mu(0) = \frac{1}{2}, \quad \nu(10) = 0 \quad \text{but} \quad \mu(10) = \frac{1}{2}.$$

Therefore,

$$\log\left(\frac{\mu(0)}{\nu(0)}\right) = \log\left(\frac{1/2}{0}\right) = +\infty, \quad \log\left(\frac{\mu(10)}{\nu(10)}\right) = \log\left(\frac{1/2}{0}\right) = +\infty.$$

Thus,

$$\text{KL}(\mu\|\nu) = +\infty.$$

In contrast, one can show that $W_1(\mu, \nu) = 1$. We skip the detailed computation but intuitively:

- The mass at 0 must move to 1: cost = $\frac{1}{2} \cdot 1$.
- The mass at 10 must move to 11: cost = $\frac{1}{2} \cdot 1$.
- The total cost is therefore 1.

3.2 Wasserstein-1 distance

Among the family of Wasserstein distances, the $p = 1$ case – often called the Earth Mover’s Distance – stands out because it admits the elegant Kantorovich–Rubinstein dual as a supremum over 1-Lipschitz functions and corresponds exactly to the intuitive “earth-mover” work cost, making it a linear-program that is robust to outliers and statistically natural. In the following, we focus on the Kantorovich–Rubinstein dual interpretation.

Theorem 2 (Kantorovich–Rubinstein duality). *The Wasserstein-1 distance given by the optimal transport cost*

$$W_1(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \sum_{x, y} \|x - y\| d\gamma(x, y) \quad (2)$$

admits the dual characterization:

$$W_1(\mu, \nu) = \sup_{\text{Lip}(f) \leq 1} \int f d\mu(x) - \int f d\nu(y).$$

Proof. Step 1: reformulating the objective

We first integrate the $\gamma \in \Gamma(\mu, \nu)$ constraint from Eq. (2) into the objective itself as follows:

$$W_1(\mu, \nu) = \inf_{\gamma} \int \|x - y\| d\gamma(x, y) + \sup_{f, g} \left[\int f(x) d\mu(x) + \int g(y) d\nu(y) - \int (f(x) + g(y)) d\gamma(x, y) \right] \quad (3)$$

where f and g are absolutely integrable functions w.r.t. μ and ν respectively. Eq. (3) encodes the $\gamma \in \Gamma(\mu, \nu)$ constraint because the supremum is 0 when $\gamma \in \Gamma(\mu, \nu)$ and is $+\infty$ otherwise. This is because

$$\int (f(x) + g(y)) d\gamma(x, y) = \int f(x) d\mu(x) + \int g(y) d\nu(y) \quad \text{if } \gamma \in \Gamma(\mu, \nu),$$

and would cancel the other terms. In any other case, f and g can be suitably chosen such that the supremum becomes $+\infty$.

We can now take the $\sup_{f, g}$ outside because the first term does not depend on f and g ,

$$W_1(\mu, \nu) = \inf_{\gamma} \sup_{f, g} \left[\int (\|x - y\| - (f(x) + g(y))) d\gamma(x, y) + \int f(x) d\mu(x) + \int g(y) d\nu(y) \right].$$

By the minimax theorem, we can replace the $\inf \sup$ with a $\sup \inf$ (this only applies under certain conditions such as compactness of the set $\mathcal{X}$), therefore

$$W_1(\mu, \nu) = \sup_{f, g} \inf_{\gamma} \left[\int (\|x - y\| - (f(x) + g(y))) d\gamma(x, y) + \int f(x) d\mu(x) + \int g(y) d\nu(y) \right].$$

Then we note that

$$\inf_{\gamma} \int (\|x - y\| - (f(x) + g(y))) d\gamma(x, y) = \begin{cases} 0 & \text{if } \|x - y\| \geq f(x) + g(y) \\ -\infty & \text{else.} \end{cases}$$

This can be transformed as a constraint in the equation as follows:

$$W_1(\mu, \nu) = \sup_{f, g} \left[\int f(x) d\mu(x) + \int g(y) d\nu(y) \right] \quad \text{subject to } f(x) + g(y) \leq \|x - y\|. \quad (5)$$

Step 2: c-transform

How do we now find such f and g that maximize the r.h.s. of Eq. (5)?

Let's assume that we have a function f and we want to find the optimal g corresponding to f that achieves the supremum in Eq. (5). We know that

$$\forall x, y, \quad f(x) + g(y) \leq \|x - y\|.$$

We can write this as follows

$$g(y) \leq \inf_x \{\|x - y\| - f(x)\}, \quad (4)$$

because if $g(y) \leq \|x - y\| - f(x) \forall x, y$, then it must in particular hold for the x that minimizes the right-hand side (which also guarantees it for all other x 's). Since $g(y) \leq \inf_x \{\|x - y\| - f(x)\}$, the best we can do to maximize the r.h.s. in Eq. (5) is to set

$$g(y) = \inf_x \{\|x - y\| - f(x)\}.$$

Eq. (4) gives us a function which is called the c -transform of f and is often denoted by f^c ,

$$f^c(y) = g(y) = \inf_x \{\|x - y\| - f(x)\}.$$

It can be shown that $f^{cc} = f$. We can now write Eq. (5) as

$$W_1(\mu, \nu) = \sup_f \left[\int f d\mu(x) + \int f^c d\nu(y) \right]. \quad (5)$$

Step 3: Finding an expression for f^c

The above duality holds for any arbitrary cost $c(x, y)$, not just for $c(x, y) = |x - y|$. In the case of $|x - y|$, let's derive the form of the dual problem (Eq. 5) when f is 1-Lipschitz.

When f is 1-Lipschitz, f^c is 1-Lipschitz too. This is true because for any given x ,

$$f^c(y; x) = \|x - y\| - f(x)$$

is 1-Lipschitz and therefore the infimum of the r.h.s.

$$f^c(y) = \inf_x \{ \|x - y\| - f(x) \}$$

is 1-Lipschitz.

Since f^c is 1-Lipschitz, for all x and y we have

$$\begin{aligned} |f^c(y) - f^c(x)| &\leq 1 \cdot \|y - x\|, \\ \implies -1 \cdot \|x - y\| &\leq f^c(y) - f^c(x) \leq 1 \cdot \|x - y\|, \\ \implies -f^c(x) &\leq \|x - y\| - f^c(y). \end{aligned} \quad (6)$$

Since Eq. (6) holds for all x and y ,

$$\begin{aligned} \implies -f^c(x) &\leq \inf_y \{ \|x - y\| - f^c(y) \}, \\ \implies -f^c(x) &\leq \underbrace{\inf_y \{ \|x - y\| - f^c(y) \}}_{f^{cc}(x)} \leq -f^c(x). \end{aligned} \quad (7)$$

The right inequality follows by choosing $y = x$ in the infimum. We know that $f^{cc} = f$, so $-f^c(x) = f(x)$ for Eq. (7) to hold.

Substituting $f^c = -f$ into Eq. (5), we obtain

$$\sup_{\text{Lip}(f) \leq 1} \int f d\mu(x) - \int f d\nu(y),$$

which is the dual form of the 1-Wasserstein distance. □

3.3 Closed-form solutions

There are only a few cases where the Wasserstein distance has a closed-form solution. We discuss two special such cases next.

1. Point Masses (Dirac Measures) If

$$\mu_1 = \delta_{a_1}, \quad \mu_2 = \delta_{a_2},$$

are Dirac measures at $a_1, a_2 \in \mathbb{R}^n$, then for any $p \geq 1$,

$$W_p(\mu_1, \mu_2) = \|a_1 - a_2\|_2.$$

If $\mu_1 = \delta_{a_1}$, $\mu_2 = \delta_{a_2}$ are Dirac measures at $a_1, a_2 \in \mathbb{R}^n$, then any coupling γ between them is a probability measure on $\mathbb{R}^n \times \mathbb{R}^n$ whose marginals satisfy

$$\gamma(A \times \mathbb{R}^n) = \delta_{a_1}(A), \quad \gamma(\mathbb{R}^n \times B) = \delta_{a_2}(B).$$

These two constraints force

$$\gamma = \delta_{(a_1, a_2)},$$

the unit mass at the single point (a_1, a_2) . Hence

$$W_p(\mu_1, \mu_2)^p = \int_{\mathbb{R}^n \times \mathbb{R}^n} \|x - y\|^p d\gamma(x, y) = \|a_1 - a_2\|^p,$$

and therefore

$$W_p(\mu_1, \mu_2) = \|a_1 - a_2\|_2.$$

2. Multivariate Gaussians ($p = 2$) Let

$$\mu_i = \mathcal{N}(m_i, C_i), \quad i = 1, 2,$$

be non-degenerate Gaussian measures in $\mathbb{R}^n$. Then

$$W_2(\mu_1, \mu_2)^2 = \|m_1 - m_2\|_2^2 + \operatorname{tr}\left(C_1 + C_2 - 2(C_2^{1/2} C_1 C_2^{1/2})^{1/2}\right),$$

and the optimal transport map is the affine map

$$T(x) = m_2 + C_2^{1/2} (C_2^{1/2} C_1 C_2^{1/2})^{-1/2} C_2^{1/2} (x - m_1).$$

The proof of this proposition is left as an exercise.